

Powershell Automation, Enumeration Tooling, and Windows Investigation Workflow

These notes were written by Kevin Beideman. The recommended time to understand and practice these concepts is 3 hours.

Powershell Deep Dive

A Powershell object is a structured piece of data that contains properties and methods. When you run the command: **Get-Process**, Powershell does not simply return text. It returns Process objects. Each process object contains properties like: ProcessName, ID, CPU, WorkingSet, and Threads.

Powershell is different from CMD because: CMD works mostly with text and Powershell works with objects.

Review: Remember that a Powershell pipeline is a system that sends the output of one command directly into another command. The pipe symbol is: |

Example: **Get-Process | Sort-Object CPU**

Get-Process creates process objects. The pipe symbol sends those objects into the next command. Sort-Object CPU sorts the objects by the CPU property.

Understand: CMD: text-based output, Powershell: object-based output.

Run: **Get-Process | Get-Member**

Get-Member is a command used to inspect an object. It tells you: what type of object something is, what properties it has, and what methods / actions it can perform. So, Get-Process |

Get-Member: first creates process objects for running processes, and sends it to Get-Member which inspects the objects and shows the aforementioned info.

Run: **Get-Process | Sort-Object CPU -Descending**. If you want to do Ascending, just say **Sort-Object CPU** because the default is Ascending order.

Run: **Get-Process | Where-Object {\$_.CPU -gt 100}**. In this command, \$_ stands for: the current object being processed. Where-Object is a command used to filter objects. "Only keep objects that match this condition."

Run: **Get-Process | Select-Object Name, CPU, WorkingSet**. Then: **Get-Service | Select-Object Name, Status**

Building First Enumeration Toolkit

An enumeration toolkit is a collection of commands, scripts, or tools used to gather information about a system, network, user, or environment. Reminder: enumeration means systematically discovering information.

Objective: Build a reusable Windows enumeration toolkit.

Script 1: Process Enumeration

Inside this file: lists top CPU Processes, lists top RAM processes, exports CSV, includes timestamp, and includes hostname. Also, include process count and currently running Powershell processes.

Script 2: Network Enumeration

Inside this file: export active TCP connections, listening ports, local IP addresses, DNS servers, and routing table

Script 3: System Enumeration

Export: hostname, OS version, uptime, logged-in user, running services, local admins